Artea

Coupon Campaign Analysis

Presented by: Team 12
Jaya Sruthi Perikala | Angie Pang |
Andrea Pan | Carson Pimental

01 | Company Background

About Artea

- Online retailer specializing in handmade clothing and accessories
- Strong Engagement
  - High time on site
  - Healthy referral rates
- Need to translate engagement → purchases

87 % of visitors never make a purchase

~5000 users in A/B experiment

20 % coupon offered to treatment group

The Experiment

- **Target Pool**
  - ~5000 visitors who browsed in past 2 month with no purchase
- **Random Split**
  - Half randomly assigned to treatment (coupon) half control (no coupon)
- **The Offer**
  - Non transferable 20% discount coupons that are valid for a month
- **Data Tracked**
  - Transactions, revenues, browsing behavior, acquisition channel, past purchases

02 | A/B Testing

Core Problem

Artea has healthy engagement but conversions. Sending coupons to everyone is risky and reduce profit and can train customers to wait for deals.

Solution

Use A/B Testing to isolate effects of the coupon and filter out browsers that might have purchased anyways.

Causual Isolation

- Controls for browsing behavior, channel, and purchasing history
- Correlation ≠ Causation

Unbiased Groups

- Treatment and control are statiscally equivalent at baseline
- No selection bias

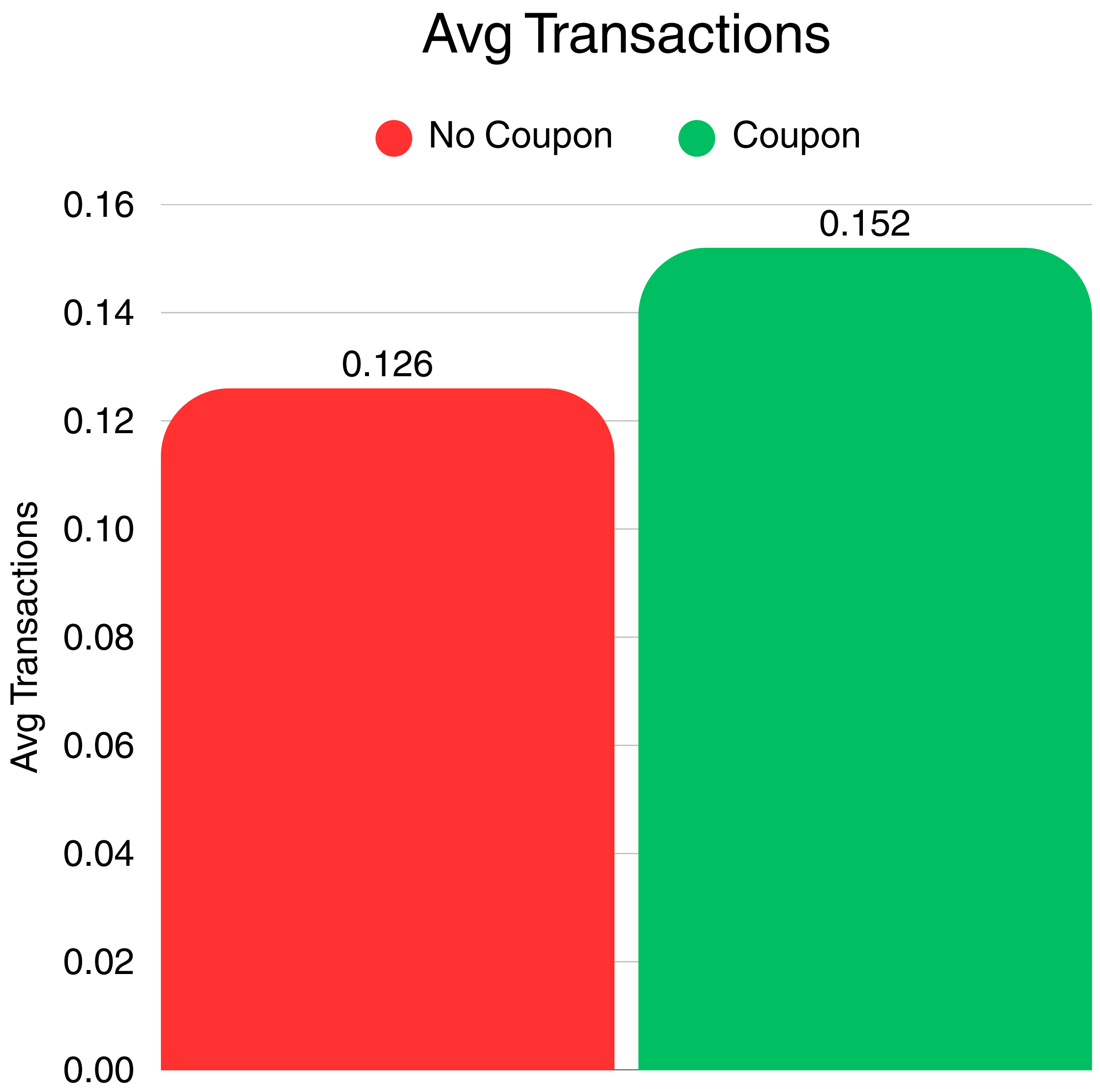
Avoid Costly Errors

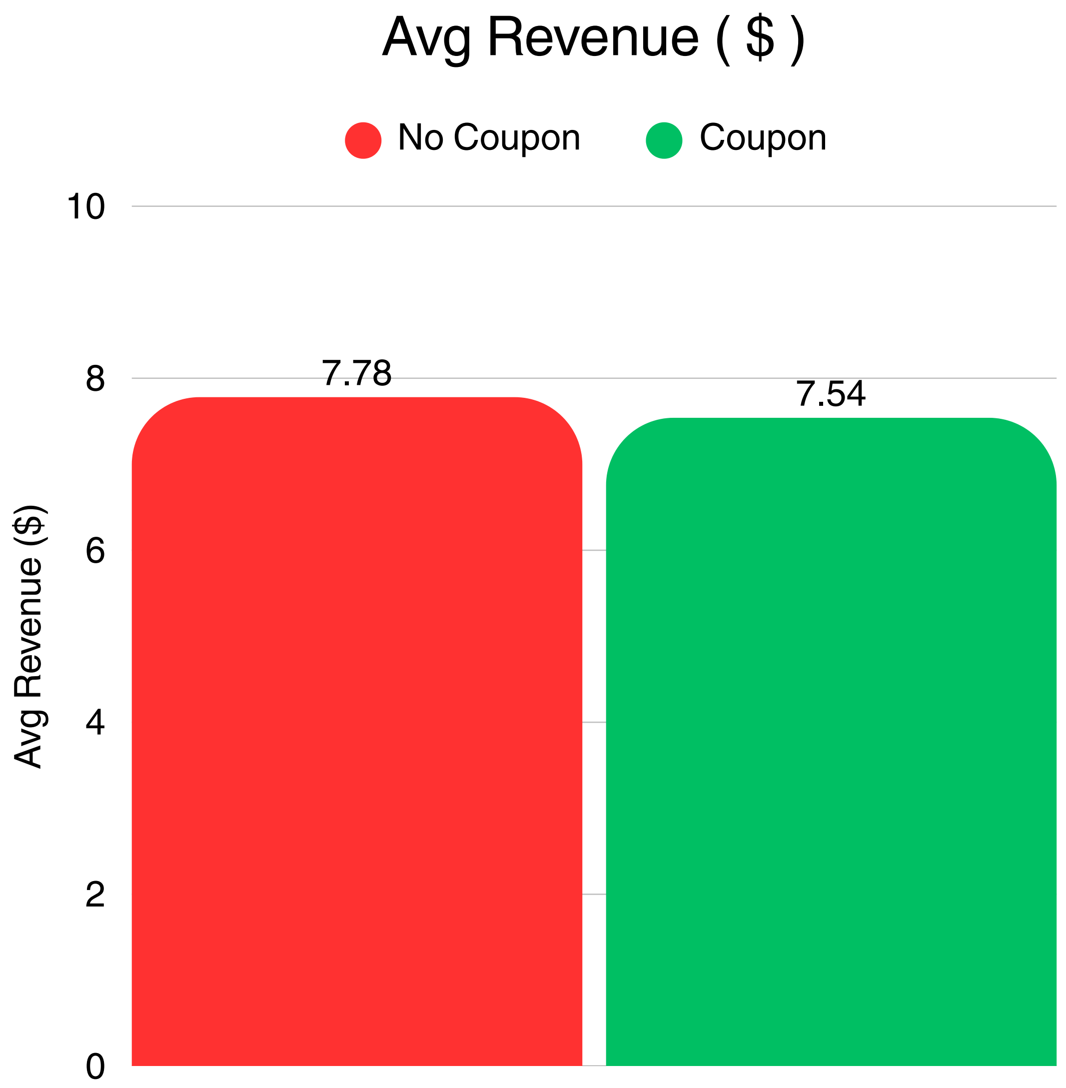
- Sending Coupons to everyone without evidence risk margins erosion at scale

Decision Confidence

- Measures the true incremental impact before fully committing to rollout

03 | Experiment Results





Transactions Increased

- +66 transactions for users with coupon
- p-value: 0.0273 → Significant
- lift of +0.0262 per user

The coupon successfully encouraged more purchases.

Revenue Decreased

- No Coupon led by $573
- p-value: 0.72 → Not Significant
- Loss of $0.24 per user on average

20% discount erased overall volume increase → Blanket Coupon doesn’t work.

04 | Analysis Approach

Goal: Identify which customers generate genuine incremental revenue from the coupon, not just those who would have bought anyway

Step-1:

Ran OLS regression on A/B test data with interaction terms to identify which customer attributes amplify coupon effectiveness

Step-2:

Used significant interaction terms to define customer segments

Step-3:

Computed uplift within each segment for average revenue and average transactions

Uplift = Average Rev/Trans (Treat) – Average Rev/Trans(Control)

Step-4:

Selected segments with positive transaction and revenue uplift

Step-5:

Applied targeting rule to Next_Campaign dataset to estimate expected outcomes

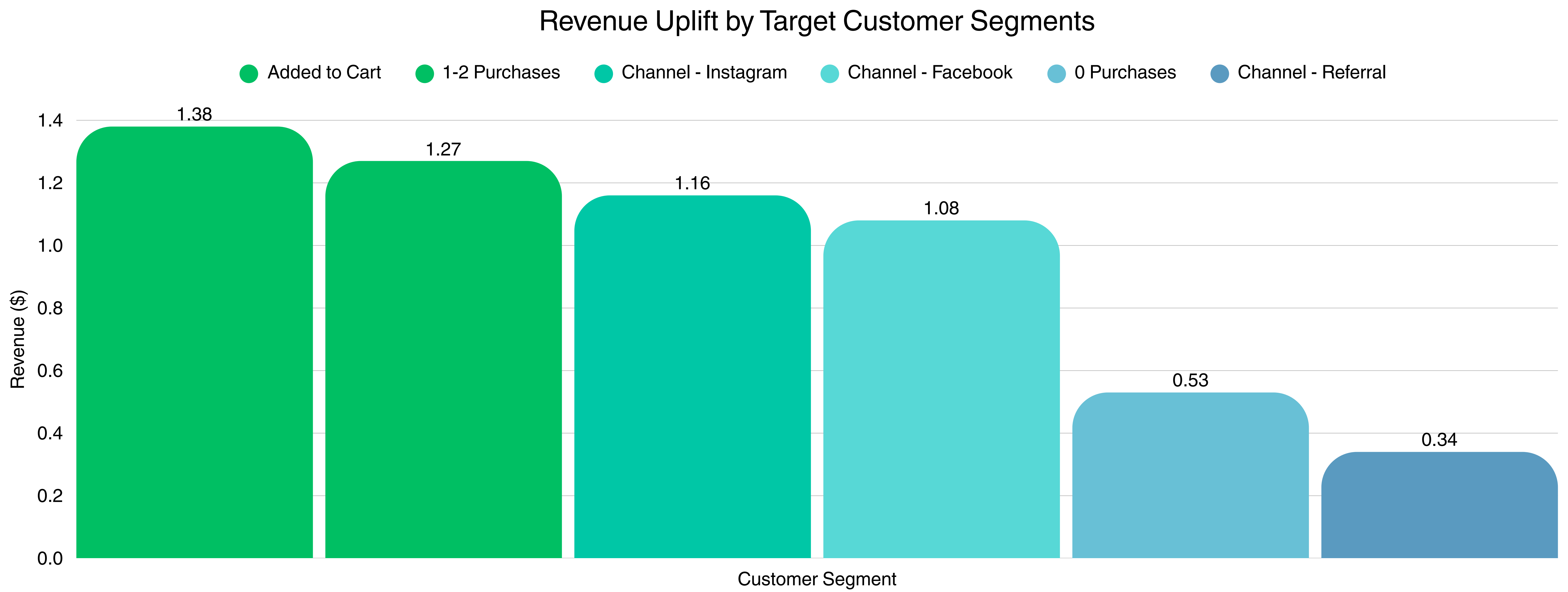
05 | Regression Findings & Targeting Rule

Significant Interaction Terms:

Variable	Coefficient	Direction	Interpretation
Shopping Cart	7.69	Positive (p=0.021)	Cart adders need a small nudge to convert
Past Purchases	-1.27	Negative (p=0.000)	Loyal customers buy anyway, discount erodes margin
Channel Acq: Instagram Channel	3.11	Positive (p=0.033)	Among all channels, users acquired through Instagram are more promotion-responsive compared to the baseline channel ‘Google’

Targeting Rule → Added to cart AND 0-2 past purchases AND acquired via Facebook, Instagram, or Referral

06 | Uplift by Segment



- Added to Cart: highest revenue uplift (+$1.38 per customer)
- 1-2 Past Purchases: strongest uplift among purchase history segments (+$1.28)
- Google and Other acquired customers show negative uplift — exclude from targeting

07 | Expected Outcomes & Risks

Metric	Value
Customers Targeted	668 (11.13%)
Expected incremental transactions	51.44
Expected incremental revenue	$922.51
Expected total revenue	$9,704.70

The incremental figures represent the additional transactions and revenue generated specifically because of the coupon, while total revenue includes the baseline purchases that would have occurred regardless.

High

Generalization of A/B Results

The targeting rule is built on a single experiment. Customer behavior, traffic sources, and purchasing patterns may shift, making the uplift estimates less reliable for future campaigns.

Med

Promotion Dependency

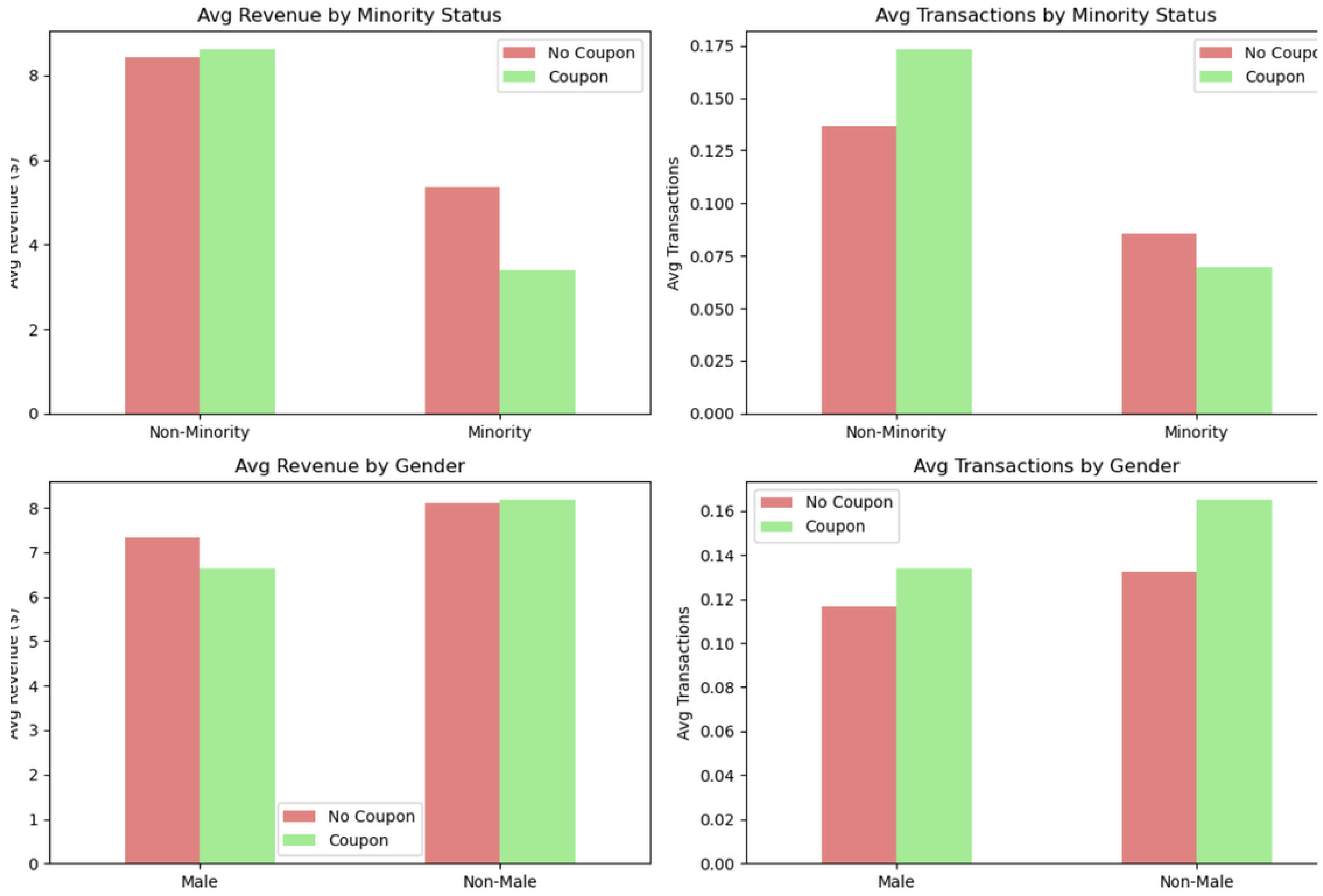
Repeated discount campaigns may train customers to only purchase when a coupon appears → erosion of full-price conversion rate and profit margins over time

Low

Fairness and Bias

Acquisition channel may correlate with demographic characteristics, meaning the targeting rule could systematically exclude certain customer groups, creating ethical and legal exposure beyond what the data reveals.

08 | Demographical Data



Interaction Term	P-Value	Interpretation
test_coupon:minority	0.639	Not Significant
test_coupon:non_male	0.164	Not Significant

Conclusion

The large p-values for the non_male and minority interaction terms indicate no significant differences in coupon effect across these groups, so the targeting policy **should not change** based on demographics.

09 | Ethical Concerns

Risk of Discrimination

Using demographic attributes in decision-making could unintentionally result in unequal treatment of certain customer groups.

Privacy Concerns

Collecting sensitive demographic information may raise concerns if customers are unaware of how their data is being collected or used.

Reputational Risk

If demographic targeting appears biased or unfair, it could lead to public criticism and damage the company's reputation.

Path Forward

User behavioral, not demographic, signals for targeting. Publish a clear data use policy. Conduct for regular fairness audits. Prioritize opt-in data collection.

THANK YOU